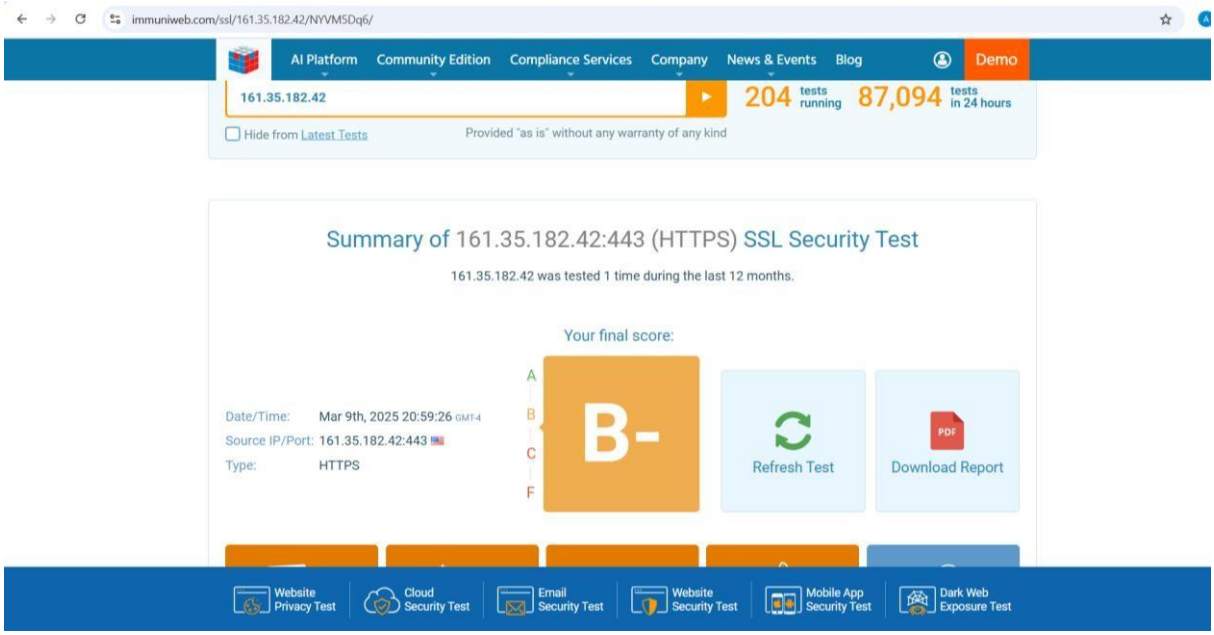
Akshaya Konduru
801456250

Project Post Quantum OpenSSL

Apache server

Apache server being tested (using immuniweb website).



Connection to the OQS test server using curl.

Akshaya-Konduru - DigitalOcean Droplet Web Console - Google Chrome

cloud.digitalocean.com/droplets/481801298/terminal/ui/?os_user=root

```
100  1809  100  1809    0     0   6062      0 --:--:-- --:--:-- --:--:--  6070
* Connection #0 to host test.openquantumsafe.org left intact
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/curl -v --curves X25519MLKEM768 --cacert $BUILD_DIR/ca.cert https://test.openquantumsafe.org:6002
* Host test.openquantumsafe.org:6002 was resolved.
* IPv6: (none)
* IPv4: 158.177.128.14
*   Trying 158.177.128.14:6002...
* ALPN: curl offers http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
*  CAfile: /root/build/ca.cert
*  CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519MLKEM768 / id-ecPublicKey
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: CN=test.openquantumsafe.org
*  start date: Jan  8 11:59:44 2025 GMT
*  expire date: Jan  8 11:59:44 2026 GMT
*  subjectAltName: host "test.openquantumsafe.org" matched cert's "test.openquantumsafe.org"
*  issuer: CN=oqstest_intermediate_ecdsap256
*  SSL certificate verify ok.
*   Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA256
*   Certificate level 1: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (4096/152 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to test.openquantumsafe.org (158.177.128.14) port 6002
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test.openquantumsafe.org:6002
> User-Agent: curl/8.13.0-DEV
> Accept: */*
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
```

Server Connection

Akshaya-Konduru - DigitalOcean Droplet Web Console - Google Chrome

cloud.digitalocean.com/droplets/481801298/terminal/ui/?os_user=root

```
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
< HTTP/1.1 200 OK
< Server: nginx/1.27.3
< Date: Mon, 10 Mar 2025 01:40:02 GMT
< Content-Type: text/html
< Transfer-Encoding: chunked
< Connection: close
<
<!DOCTYPE html>
<html>
<head>
<title>Open Quantum Safe interop test server for quantum-safe cryptography</title>
</head>
<body>
<h1 align=center>
Successfully connected using
ecdsap256-X25519MLKEM768!
</h1>

Client-side KEM algorithm(s) indicated:
X25519MLKEM768
</body>
</html>

* shutting down connection #0
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl req -x509 -new -newkey falcon512 -keyout falcon512_CA.key -out falcon512_CA.crt -nodes -subj "/CN=test CA" -days 365
nfig $OPENSSL_CONF
-----
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl genpkey -algorithm falcon512 -out dilithium3_srv.key
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl req -new -newkey falcon512 -keyout falcon512_srv.key -out falcon512_srv.csr -nodes -subj "/CN=test server" -config $O
SSL_CONF
-----
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl x509 -req -in falcon512_srv.csr -out falcon512_srv.crt -CA falcon512_CA.crt -CAkey falcon512_CA.key -CAcreateserial -
s 365

Certificate request self-signature ok
subject=CN=test server
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl s_server -cert falcon512_srv.crt -key falcon512_srv.key -www -tls1_3 -groups frodo640shake
Using default temp DH parameters
ACCEPT
```

Client Connection

Akshaya-Konduru - DigitalOcean Droplet Web Console - Google Chrome

cloud.digitalocean.com/droplets/481801298/terminal/ui/

```
*** System restart required ***
Last login: Mon Mar 10 00:43:33 2025 from 162.243.188.66
root@Akshaya-Konduru:~# cd curl
root@Akshaya-Konduru:~/curl# export WORKSPACE=/root
root@Akshaya-Konduru:~/curl# export BUILD_DIR=$WORKSPACE/build
root@Akshaya-Konduru:~/curl# export OPENSSL_CONF=$BUILD_DIR/ssl/openssl.cnf
root@Akshaya-Konduru:~/curl# export OPENSSL_MODULES=$BUILD_DIR/lib
root@Akshaya-Konduru:~/curl# $BUILD_DIR/bin/openssl s_client -groups frodo640shake
Connecting to ::1
CONNECTED(00000003)
depth=0 CN=test server
verify error:num=20:unable to get local issuer certificate
verify return:1
depth=0 CN=test server
verify error:num=21:unable to verify the first certificate
verify return:1
depth=0 CN=test server
verify return:1
---
Certificate chain
 0 s:CN=test server
   i:CN=test CA
   a:PKEY: UNDEF, 128 (bit); sigalg: falcon512
   v:NotBefore: Mar 10 01:43:10 2025 GMT; NotAfter: Mar 10 01:43:10 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIG6jCCBEegAwIBAgIUdC9AtxK5nn+bvwXmEk2Yfzz1iyswBwYFK84PAwswEjEQ
MA4GA1UEAwwHdGVzdCBDQTAeFw0yNTAzMTAwMTQzMTBaFw0yNjAzMTAwMTQzMTBa
MBYxFDASBgNVBAMMC3Rlc3Qgc2VydmVyMIIDjzAHBgUrzg8DCwOCA4IACWO6MLis
aFEplocjIxE+sb6D8qSYf6lUjLVvCLe+2tJb6EzfP+wTcIDasnCJGsnOjVwcunLQ
2mDppwlbZBmlWyXFW/G6JMQIgg54x13/urHCxbBSaXgJRHSjpzsj6ksVaROXfVaq
Vb7jT6YYEAgYlOZeWFkZDdJAWkaP8mwjSbd9fUBM6cte1glmo5E4igbhEcXBR1sa
+0o9naifahWrWkpcEV0BCIdVwTp+ughxF9KkaVTIlW5UKc0Ms/jqaCF2nKfYiOaI
oRuZcdu0aaReWpfHM2WttPo+Zj8KIFuhal86uLi60caRvVR8BjaP1YoTfbFrh4v3
G05HgKLX3VZqdviBQVYqNt3zUNBvRaWxUMwQJEMBBbHjoct9yXEl74+AMIwwZvEk
Ab5p7/pcFnt8NQlyIA7oBnmZGfWJ85NZcGu0q2OA3RwDoltgRCDguCCiF20ZpnKv
bnScpdOyFt1Ure22dg5dgL80R9L+QWwSNi7HuS7ZceHNoBjK1jPCRpBY3xKyHXpi
rYU8oFyfYeSsCVOil61anJwFbgA1J1z/chh+aV0ADELd1ZuK0hx/xarmzIrcUsFB
g2weQ0dYsvtqqqKl/GA4v5dYAvovaNgERlvc4ZvjTHwcxq/hVIxakyr4lb+bHkN8
4TgPWU2t5mzyi4QJgZASMuhXhmy1Z6nGkzZYvR80eZeHIW#GRVha0VVYJsI/iqFo
```